MIDTERM 1
MATH 54

1. (a) Are vectors

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \ \mathbf{u}_2 = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}, \ \mathbf{u}_3 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

linearly independent in $\mathbb{R}^3$?

(b) Find all values of h for which $\begin{bmatrix} 2 \\ 3 \\ h \end{bmatrix}$ lies in the $\mathrm{Span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$.

2. (a) Compute the determinant of the matrix

$$A = \begin{bmatrix} 2 & -1 & 0 \\ 0 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}.$$

(b) Is A invertible? If yes, compute the inverse matrix.

3. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation such that

$$T(\mathbf{e}_1) = \mathbf{e}_1 + \mathbf{e}_2, \ T(\mathbf{e}_2 - \mathbf{e}_1) = \mathbf{e}_1 - \mathbf{e}_2.$$

(a) Compute $T(\mathbf{e}_2)$ and find the standard matrix of T.

(b) Find the area of the image of the unit circle under the mapping $\mathbf{x} \mapsto T(\mathbf{x})$.

4. Consider the set S of all 2×2 matrices A such that $A^T = A$.

(a) Check that S is a subspace in the vector space of all 2×2 matrices.

(b) Find a basis of S.